	Qwen3-8B (T=1.0)			Qwen3.6-27B (T=1.5)			Gemma-2-9B (T=1.5)			Gemma-3-27B (T=1.3)		
k	Acc	ECE	AUROC	Acc	ECE	AUROC	Acc	ECE	AUROC	Acc	ECE	AUROC
3	.285	.240	.808	.107	.313	.735	.112	.307	.698	.298	.223	.814
5	.334	.131	.823	.138	.204	.766	.141	.198	.745	.357	.110	.824
10	.371	.069	.835	.173	.131	.797	.184	.101	.739	.404	.097	.853
20	.402	.057	.829	.206	.103	.812	.214	.054	.748	.422	.117	.876

Table 10: Bootstrap sample-count ablation: mode frequency over the first k of the 20 saved draws, at the ECE-optimal temperature of the original grid for each oracle. Most of the calibration and ranking gain is in place by k=10 (on Gemma-3-27B the ECE minimum sits at k=10); k is the main cost lever of the bootstrap.

Word	Qwen3-8B	Qwen3.6-27B	Gemma-2-9B	Gemma-3-27B
moon	.827	.380	.623	.467
snow	.727	.413	.320	.487
jump	.627	.240	.633	.553
smile	.610	.240	.623	.423
ship	.580	.100	.263	.473
chair	.557	.190	.133	.513
flag	.523	.097	.513	.477
salt	.477	.277	.060	.527
green	.473	.450	.213	.497
dance	.350	.197	.043	.470
book	.337	.267	.153	.323
gold	.327	.247	.047	.300
cloud	.307	.163	.073	.593
flame	.297	.147	.070	.170
leaf	.240	.050	.087	.373
blue	.180	.500	.157	.560
wave	.170	.110	.000	.467
clock	.170	.157	.277	.277
song	.150	.100	.000	.047
rock	.120	.170	.087	.587

Table 11: Per-word accuracy on Bootstrap T=1.0 for all four oracles, sorted by Qwen3-8B accuracy. Per-word accuracy spans an order of magnitude on every oracle.

2.8.3, Python 3.13.

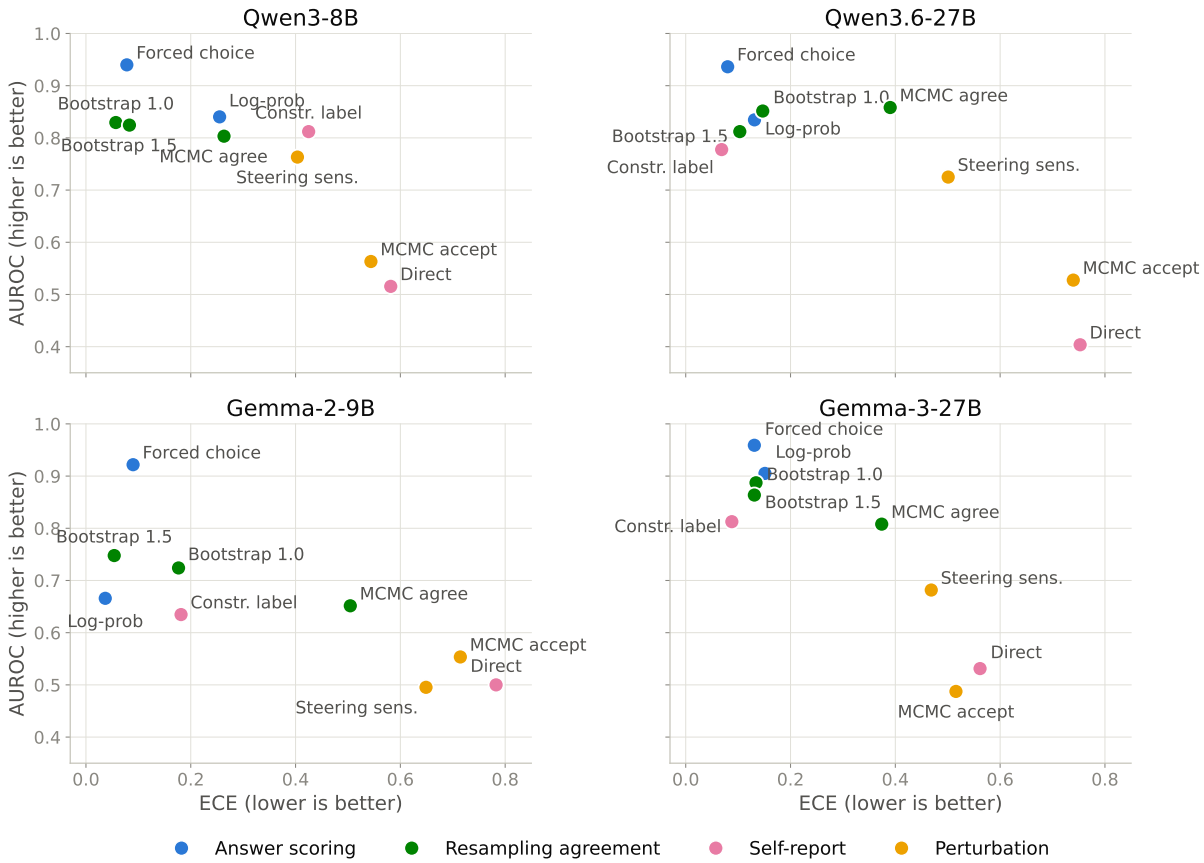


Figure 2: Calibration versus ranking on all four oracles. Each point is one confidence method; the best corner is top left. Colour groups methods by mechanism; every point is labeled directly. Forced choice and the bootstrap family occupy the good corner on every oracle; direct numeric self-report and raw MCMC acceptance sit in the high-ECE, near-chance corner on every oracle.

	Qwen3-8B				Qwen3.6-27B			
Method \ N	2	5	10	20	2	5	10	20
Bootstrap T=1.0 ECE	0.213	0.045	0.049	0.057	0.130	0.139	0.155	0.147
Bootstrap T=0.7 ECE	0.139	0.108	0.080	0.097	0.194	0.196	0.230	0.218
Bootstrap T=1.5 ECE	0.244	0.080	0.084	0.083	0.067	0.084	0.106	0.103
Log-prob (offset) ECE	0.487	0.245	0.275	0.255	0.157	0.160	0.119	0.131
MCMC agree T=0.5 ECE	0.047	0.269	0.242	0.263	0.350	0.374	0.395	0.390
Direct (numeric) ECE	0.291	0.584	0.563	0.582	0.710	0.719	0.764	0.753
Bootstrap T=1.0 AUROC	0.795	0.791	0.819	0.829	0.842	0.837	0.852	0.851
Bootstrap T=0.7 AUROC	0.783	0.801	0.815	0.830	0.854	0.847	0.871	0.863
Log-prob (offset) AUROC	0.829	0.815	0.828	0.840	0.835	0.820	0.847	0.835
MCMC agree T=0.5 AUROC	0.775	0.753	0.786	0.803	0.866	0.834	0.865	0.858

Table 12: Controlled target-set scaling on the two Qwen oracles. Bootstrap T=1.0 on Qwen3-8B is the only method whose ECE improves as N grows; most others degrade. On Qwen3.6-27B every method is roughly N-stable.

	Qwen3-8B			Qwen3.6-27B			Gemma-2-9B			Gemma-3-27B		
Method	Acc	ECE	AUROC	Acc	ECE	AUROC	Acc	ECE	AUROC	Acc	ECE	AUROC
MCMC agreement T=0.5	.413	**.263** ±.005	**.803** ±.006	.228	**.390** ±.005	**.858** ±.006	.214	**.504** ±.006	**.652** ±.008	.435	**.374** ±.005	**.808** ±.005
Steering sensitivity	.418	.404 ±.006	.763 ±.005	.232	.501 ±.005	.725 ±.007	.208	.649 ±.006	.495 ±.008	.433	.469 ±.006	.682 ±.005
MCMC accept T=0.125	.415	.544 ±.006	.563 ±.005	.234	.740 ±.006	.528 ±.004	.217	.714 ±.005	.553 ±.006	.436	.516 ±.007	.487 ±.007

Table 13: The three confidence readouts built for the steered setting, n=6,000 samples per row per oracle, at the best temperature of each grid. ECE and AUROC carry ± one bootstrap standard error (1000 resamples); best ECE and AUROC per oracle among these three in **bold**. Every row here is weaker on both axes than the tuned bootstrap of table 1.

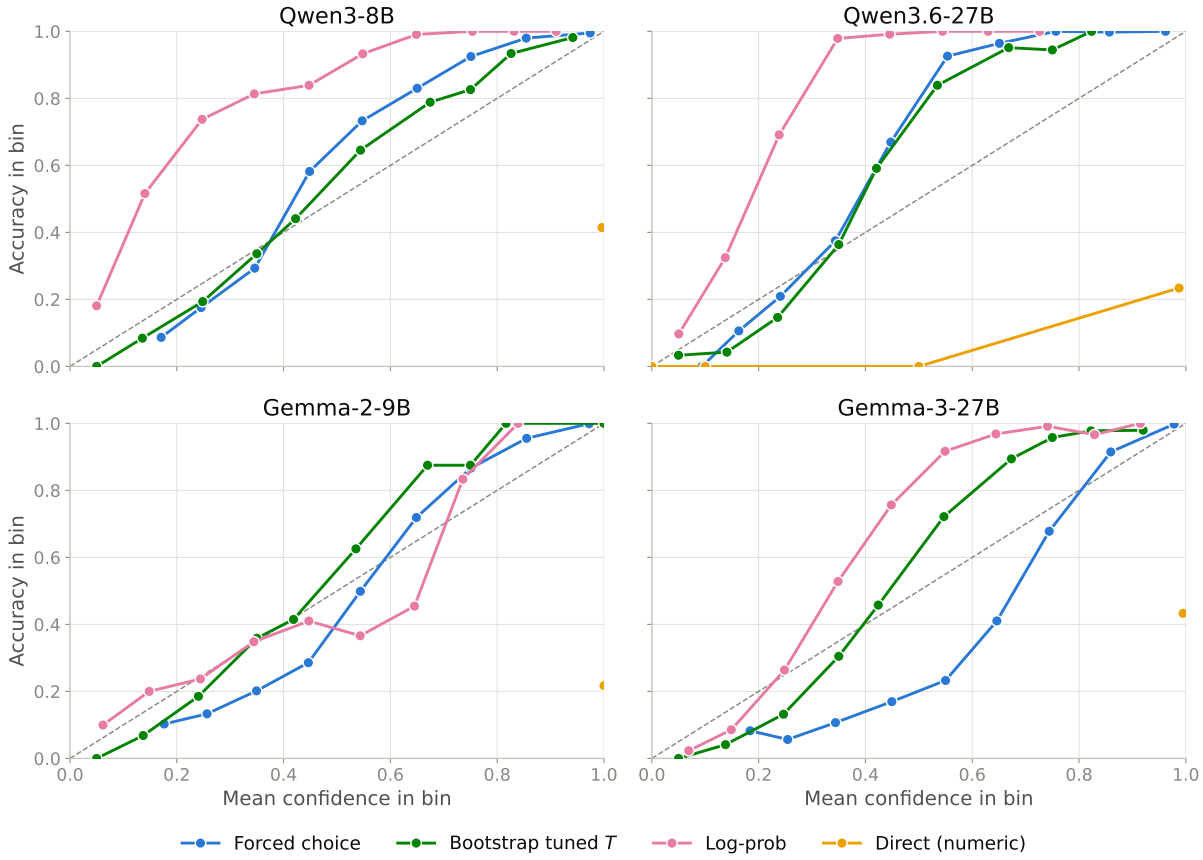


Figure 3: Reliability diagrams on all four oracles. The dashed diagonal is perfect calibration; bins with no samples are omitted. Direct numeric self-report appears as one or two points at confidence near 1 because the oracle reports the same high number on almost every sample.

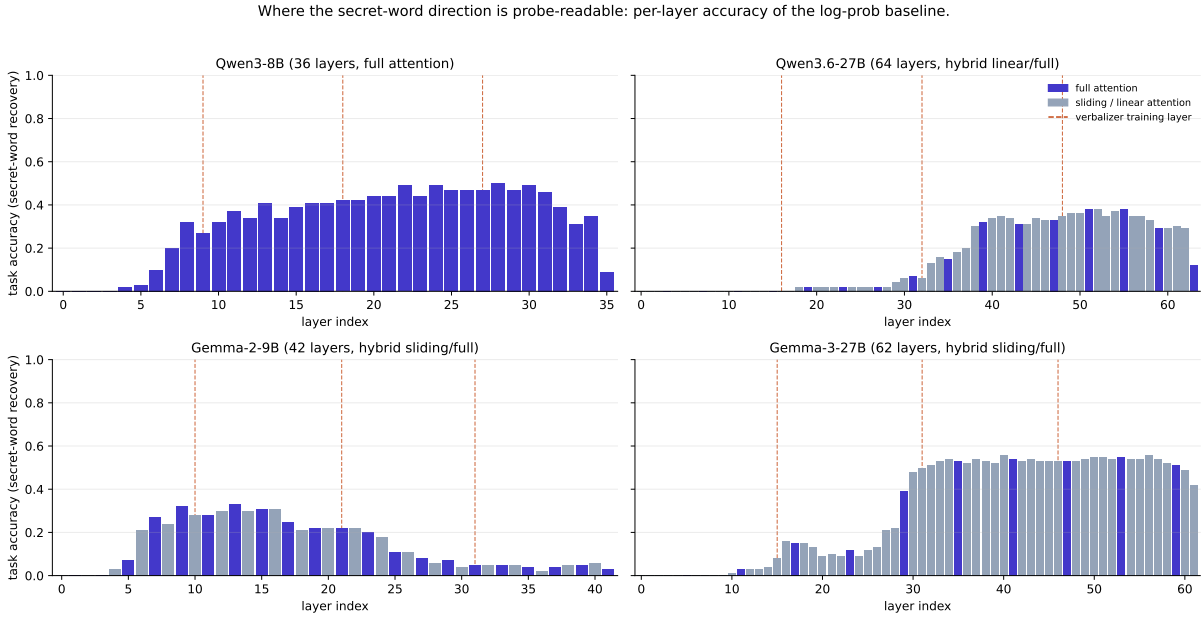


Figure 4: Per-layer log-prob accuracy on the secret-word task across all decoder layers of four oracle bases. Dashed lines mark the three layers each verbalizer was trained on; bar shading distinguishes full-attention (indigo) from sliding/linear-attention (slate) layers. Band peaks: Qwen3-8B 50% at layer 28; Qwen3.6-27B 38% at layer 51; Gemma-2-9B 33% at layer 13; Gemma-3-27B 56% at layer 40 (broad plateau $\sim$ 50–56% across layers 31–59). Gemma-2-9B reads early; the Qwens read late; Gemma-3-27B has the widest probe-readable band of the four.

		Qwen3-8B			Qwen3.6-27B		
T	α	Acc	ECE	AUR.	Acc	ECE	AUR.
.500	2	.376	.551	.601	.211	.746	.574
.250	4	.403	.547	.579	.230	.738	.547
.125	8	.415	.544	.563	.234	.740	.528

Table 14: M6 (power-sampling acceptance) temperature sweep on the two Qwen oracles; the Gemma rows are in section A. AUR.: AUROC. As T drops the chain accepts more on correct and wrong inputs alike, shrinking the gap between the two pools.

Method	Acc	ECE	Br.	AUR.
Qwen3-8B				
Bootstrap T=1.0	.402	.057	.163	.829
MCMC agree T=0.5	.413	.263	.247	.803
Qwen3.6-27B				
Bootstrap T=1.0	.225	.147	.130	.851
MCMC agree T=0.5	.228	.390	.274	.858

Table 15: Bootstrap versus power-sampling agreement at the best temperature of each. *Br.*: Brier. *AUR.*: AUROC.

		Qwen3-8B			Qwen3.6-27B		
T	α	Acc	ECE	AUR.	Acc	ECE	AUR.
.500	2	.413	.263	.803	.228	.390	.858
.250	4	.418	.408	.737	.233	.569	.767
.125	8	.415	.493	.669	.236	.668	.653

Table 16: M7 (power-sampling agreement) temperature sweep. The $\alpha = 2$ row is competitive with bootstrap on AUROC, especially on Qwen3.6-27B (AUR. 0.858), at $5\times$ the wall-clock cost and with weaker ECE.